

## Full length article

## Crowdsourced reliable labeling of safety-rule violations on images of complex construction scenes for advanced vision-based workplace safety

Yanyu Wang<sup>a</sup>, Pin-Chao Liao<sup>b</sup>, Cheng Zhang<sup>c,\*</sup>, Yi Ren<sup>d</sup>, Xinlu Sun<sup>b</sup>, Pingbo Tang<sup>a</sup><sup>a</sup> Del E. Webb School of Construction, Arizona State University, Tempe, AZ 85281, USA<sup>b</sup> Department of Construction Management, Tsinghua University, Beijing 100084, PR China<sup>c</sup> Zachry Department of Civil Engineering, Texas A&M University, College Station, TX 77843, USA<sup>d</sup> School for Engineering of Matter, Transport & Energy, Arizona State University, Tempe, AZ 85281, USA

## ARTICLE INFO

## Keywords:

Crowdsourcing  
Construction safety  
Image annotation  
Bayesian network model  
Safety inspection

## ABSTRACT

Construction workplace hazard detection requires engineers to analyze scenes manually against many safety rules, which is time-consuming, labor-intensive, and error-prone. Computer vision algorithms are yet to achieve reliable discrimination of anomalous and benign object relations underpinning safety violation detections. Recently developed deep learning-based computer vision algorithms need tens of thousands of images, including labels of the safety rules violated, in order to train deep-learning networks for acquiring spatiotemporal reasoning capacity in complex workplaces. Such training processes need human experts to label images and indicate whether the relationship between the worker, resource, and equipment in the scenes violate spatiotemporal arrangement rules for safe and productive operations. False alarms in those manual labels (labeling no-violation images as having violations) can significantly mislead the machine learning process and result in computer vision models that produce inaccurate hazard detections. Compared with false alarms, another type of mislabels, false negatives (labeling images having violations as “no violations”), seem to have fewer impacts on the reliability of the trained computer vision models.

This paper examines a new crowdsourcing approach that achieves above 95% accuracy in labeling images of complex construction scenes having safety-rule violations, with a focus on minimizing false alarms while keeping acceptable rates of false negatives. The development and testing of this new crowdsourcing approach examine two fundamental questions: (1) *How to characterize the impacts of a short safety-rule training process on the labeling accuracy of non-professional image annotators?* And (2) *How to properly aggregate the image labels contributed by ordinary people to filter out false alarms while keeping an acceptable false negative rate?* In designing short training sessions for online image annotators, the research team split a large number of safety rules into smaller sets of six. An online image annotator learns six safety rules randomly assigned to him or her, and then labels workplace images as “no violation” or “violation” of certain rules among the six learned by him or her. About one hundred and twenty anonymous image annotators participated in the data collection. Finally, a Bayesian-network-based crowd consensus model aggregated these labels from annotators to obtain safety-rule violation labeling results. Experiment results show that the proposed model can achieve close to 0% false alarm rates while keeping the false negative rate below 10%. Such image labeling performance outdoes existing crowdsourcing approaches that use majority votes for aggregating crowdsourced labels. Given these findings, the presented crowdsourcing approach sheds lights on effective construction safety surveillance by integrating human risk recognition capabilities into advanced computer vision.

## 1. Introduction

Safety inspection aims for proactive management of hazardous conditions. The goal is to achieve “zero accidents” [1]. However, the proportion of hazards identified by human inspectors has reached a

plateau [2]. Automated workplace safety monitoring and diagnosis based on computer vision and deep learning technologies could achieve reliable and real-time detection of construction workplace hazards [3–6]. A step toward automated workplace safety monitoring is to enable computer algorithms to learn the logic behind numerous safety

\* Corresponding author.

E-mail addresses: [ynwang@asu.edu](mailto:ynwang@asu.edu) (Y. Wang), [pinchao@tsinghua.edu.cn](mailto:pinchao@tsinghua.edu.cn) (P.-C. Liao), [c Zhang@tamU.edu](mailto:c Zhang@tamU.edu) (C. Zhang), [yiren@asu.edu](mailto:yiren@asu.edu) (Y. Ren), [sun-xl17@mails.tsinghua.edu.cn](mailto:sun-xl17@mails.tsinghua.edu.cn) (X. Sun), [tangpingbo@asu.edu](mailto:tangpingbo@asu.edu) (P. Tang).

<https://doi.org/10.1016/j.aei.2019.101001>

Received 12 September 2018; Received in revised form 5 August 2019; Accepted 2 October 2019

Available online 18 October 2019

1474-0346/ © 2019 Elsevier Ltd. All rights reserved.

rules through the use of example workplace images labeled by humans as containing safety-rule violation scenarios or not. A dataset consisting of images labeled as violating or following certain safety rules is essential to serve as the input of machine learning algorithms for training automated computer-vision-based workplace safety monitoring systems. However, manually labeling numerous workplace images by safety engineers was tedious and costly due to the limited availability of experienced safety engineers and huge manual image labeling workloads [4]. Even experienced engineers could only achieve 70% accuracy in labeling safety rule violations of complex scenes due to (1) distractions caused by a large number of objects in images, (2) subtle differences between safety rule definitions, and (3) misperceptions about objects and relationships in scenes [7].

Crowdsourcing is a promising alternative for labeling large amounts of images that contain scenes that comply with or violate certain safety rules [8,9]. A crowdsourcing approach invites randomly recruited online annotators to collect safety-rule violation responses of images and then aggregates their contributions into safety-rule-violation labels. Such aggregation integrates the recognition power of people without professional safety knowledge into formal safety-related reasoning capacity. Unpredictable human behavior and dynamic workspaces pose challenges of hazard detection in complex contexts, while crowdsourcing approaches are promising for addressing these challenges [8,10–12].

Image labels crowdsourced from non-professional annotators need proper “annotation aggregation” – a process of resolving conflicting information from multiple labels of an image and obtaining the final label of that image as certain violations. In other words, the aggregation model sorts out the correct labels despite various errors in the crowdsourced annotations. However, existing methods for such aggregation become unreliable when a significant portion of labels are wrong. One commonly used method for aggregating the answers of crowdsourced participants is “majority vote”; this algorithm identifies the correct answer as the answer agreed on by the most participants [13–16]. While majority votes yield accurate results in many cases, such methods often fail when a significant portion of participants are wrong [15,17,18], i.e., the majority vote mistake.

Another category of annotation aggregation methods is Expectation Maximization-Based (EM-based). EM-based methods have the similar difficulties encountered by the majority-vote methods while handling cases where significant portions of answers are wrong [19]. When over half of the workers choose the wrong answer while dealing with complex scenes, EM-based methods will likely converge to that incorrect answer. For example, Demartini et al. found that ZenCrowd, an elaborate EM-based method, cannot achieve precision higher than 85% no matter how large the number of crowd workers is [19,20]. In many cases, unreliable annotations produced by people who lack professional inspection training can cause biases in the aggregated results: many images with no safety-rule violation would be labeled as having violations, or “false alarms”.

High rates of false alarms in safety-violation labeling of photos, called “high false-positive rates,” can seriously mislead the training process of deep-learning-based computer vision algorithms. In fact, without additional field information depicting operational contexts, the use of training images with many false positives to train deep learning networks for automatic labeling of photos with safety violations will result in algorithms that have a large number of mislabels [21]. Computer vision researchers have been tackling noisy labels of training example images for a decade. Many researchers developed methods for tolerating false-negatives in the training data [22–25]. Unfortunately, these studies found that few machine learning methods have yet to overcome high false-positive rates in the training data set [26]. Therefore, reducing false positive rates of crowdsourced safety-rule violation labels, along with minimizing the increase of the false negative rates, is critical.

This paper examines a new low-cost crowdsourcing approach that

can achieve the lowest possible false alarm rates with acceptable false negative rates in labeling workplace images as containing certain safety-rule violation scenarios. The new approach aggregates the annotations contributed by ordinary people into a labeled workspace image dataset with the minimum false alarms and acceptable false negatives in the image labels of safety-rule violations. The authors expect that this workspace image dataset with crowdsourced labels can serve as a reliable training data set for deep-learning-based computer vision algorithms that can automate safety rule compliance checking based on field videos. The presented study investigates the fundamental crowdsourcing methodology in the domain of vision-based safety surveillance through the following:

- 1) Characterizing the impacts of six-rule and ten-minute long training processes on the labeling accuracy of the crowdsourced annotations, compared to published works on crowdsourced annotations without any training processes;
- 2) Examining a new crowd consensus model that automatically aggregates the violation annotation results from briefly trained annotators into image labels. The aggregated image labels have a low false positive rate and an acceptable false negative rate for training vision-based safety surveillance algorithms.

Results from an online image-based safety-violation labeling experiment showed that the proposed innovations result in a new crowdsourcing approach, one that achieved reliable safety-violation labeling results in twenty target photos when compared with the results generated by the majority votes. The Background Research section below introduces previous studies related to construction safety-rule violation and crowdsourcing techniques while highlighting their contributions to this paper. The Methodology section describes the details of the new crowdsourcing approach that integrates the short training process and the new crowd consensus model. The Validation section compares the performance of the proposed new crowdsourcing approach against the conventional majority vote approach for annotation aggregation. The performance metrics used in this comparison include true positive, true negative, false positive, false negative, and correct rate. The Discussion and Conclusion sections present the results, conclusion, and future research plans.

## 2. Background research

This section first reviews existing computer vision-based (CV-based) approaches in workplace safety. Then the authors will discuss how comprehensive datasets can improve CV-based safety surveillance approaches and the need for large training image datasets for such approaches. Lastly, this section will introduce crowdsourcing techniques as a solution to acquire large training image datasets and the challenges of achieving reliable crowdsourced image labels.

Manual safety-rule violation detection is costly, time-consuming, and error-prone [27–29]. Researchers have been trying to automate safety-rule violation detection through the use of computer vision (CV). For example, Rubaiyat et al. [30] developed CV methods for automatically detecting misses of construction helmets from the workspace images. Han and Lee [1] proposed a framework of CV-based unsafe action detection on construction sites. This research used workplace videos to reconstruct three-dimensional (3D) human skeleton models containing motion information of workers in order to detect pre-defined unsafe actions. Alongside academia, the industry was also developing CV approaches for workplace safety surveillance. A commercial solution [31], “Smartvid.io”, supported the collection, organization, and retrieval of workplace videos and can automatically label the objects appearing in a construction scene, all based on CV and audio-based speech content analysis. Microsoft Build 2017, which is the worldwide developer’s conference of Microsoft, announced a state-of-the-art Artificial Intelligence (AI) approach for workplace safety [6]. In the

announced approach, AI recognized the objects captured by a camera and can instantly notice when a non-employee is in a workplace by mistake, or identify when the positioning of tools violate safety rules.

To further improve the reasoning of task-specific or safety-rule-specific object relations of computer vision techniques, the academy and industry need a large number of images in construction workplaces that show different safety-compliance or safety-violation scenes [32–37]. Most of the existing safety-oriented CV approaches focused on detecting certain safety-related scenarios (e.g., detecting whether workers are wearing hard hats, [30,38]) or using small datasets containing less than 10 k images to train machine-learning models [39,40]. However, the reliability of such approaches in detecting object-relation safety violations in different circumstances was rather limited [4,5,41]. This limitation is because acquiring a large dataset with images labeled as violating/following different object-relation safety-rules is challenging. Liao et al. (2016) presented the observation misses in a simulated safety inspection experiment of an elevator installation project. The experiment results indicated that the inspectors could only identify 15–24% of all safety-rule violations, due to complexity in object relationship analyses. Moreover, only 62% of the identified safety-rule violations were actual violations. Such results revealed that the low hazard detection rate might result from the large cognitive load generated by triangulating enormous safety criteria and scenes [7]. Moreover, labeling a large number of images by a few professional engineers could be expensive and tedious [4]. The average salary of one safety specialist was approximately 75,000 dollars a year [42–44]. According to the author's previous study [7], using six specialists to annotate twenty images would on average require twenty minutes per person, so that annotating one image costs approximately 3.6 dollars.

Crowdsourcing is a widely used image labeling approach that can potentially address the challenge of acquiring workplace safety-oriented training datasets [36,45]. Crowdsourcing techniques make decisions based on aggregating the thoughts of agents in achieving higher decision quality than that based on the opinions of individuals [16,46]. Liu et al. proposed a crowdsourcing approach that derives timely and reliable construction activity analysis based on workplace video streams through vision-based tracking algorithms and activity identification of anonymous online non-experts [11]. Compared to the expert-based image annotation, crowdsourcing is considerably less expensive. In the authors' experiment, hiring one worker to annotate twenty images only cost 0.02 dollars and a crowdsourced image labeling result requires the annotation results of 50 workers at most. Therefore, labeling one image via crowdsourcing cost about 0.05 dollar,

which could save significant image labeling costs.

Successful applications of crowdsourcing in engineering domains also reveal the challenges of reliable crowdsourcing. Some crowdsourcing studies focused on optimizing clearly-defined objective functions in order to evaluate the effectiveness of crowdsourced responses [47]. Other research questions for which the correct or best answer was unknown, however, is more challenging. Unknown ground truths (correct labels of images) brought difficulties in aggregating the answers from participants and deriving the correct answer when sometimes most of the participants' answers were wrong [15,17,19]. Drapeau et al. [19] described many techniques that could handle cases in which the majority of votes could be wrong. The most commonly used approach is to add questions with known answers to the question pool so as to evaluate the intellectual capability and identify gaming behaviors of the participants (e.g., random selections without thinking about questions). Tournament voting [48] allowed the participants to evaluate answers from other crowdsourcing participants and vote to delete suspicious answers. MicroTalk [19] allowed the participants to debate when answering the crowdsourcing questions and thus achieve consensus. However, these approaches assumed that majority-vote mistakes were rare in crowdsourcing games, which was not always true [18]. When most participants frequently chose the wrong answer, the crowdsourced results would be severely biased [49]. Also, most of the approaches did not provide the necessary background knowledge to annotators in a short but effective training. Such training process was essential for professional safety-related interpretation tasks, although not necessary for the tasks that the common sense of most ordinary people could handle (e.g., identify buildings and people in a scene).

In summary, crowdsourcing approaches can either derive group decisions on true positive answers or eliminate true negative evaluations. To fully utilize the advantage of the integration of humans into automation, crowdsourcing-based safety-rule violation detection requires an effective approach that improves the safety-rule violation capability of ordinary people through a training process and then integrates the annotations from ordinary people in an effective way to overcome the limitations of majority votes.

### 3. Methodology

The goal of the developed new crowdsourcing method is to achieve more reliable safety-rule violation labeling of workplace images to reduce high false-positive rates while keeping acceptable false negative rates. Such fundamental improvements of labeling reliability can

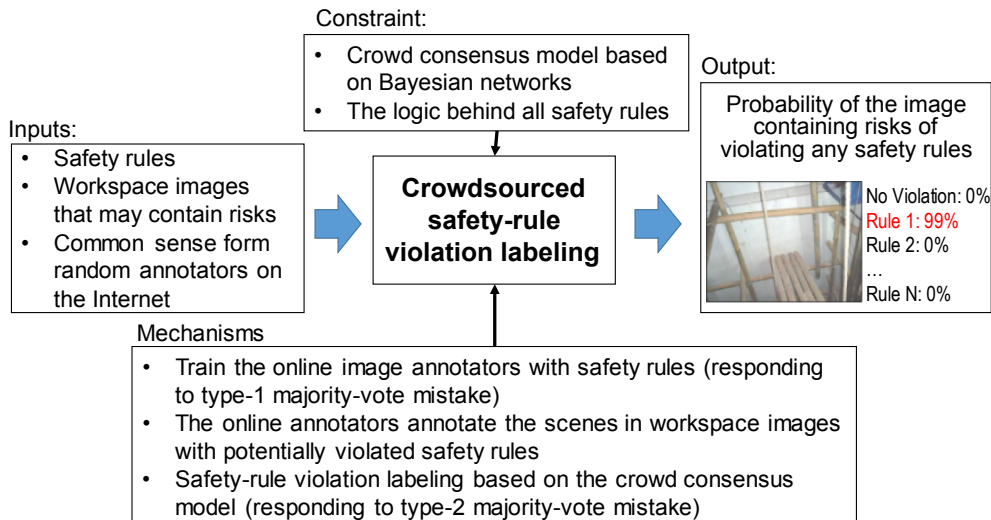


Fig. 1. IDEFO model (a method designed to model the decisions, actions, and activities of an organization or system) of crowdsourcing-based safety-rule violation labeling.

provide CV algorithms a better basis to learn correct interpretation of images from a human. Fig. 1 shows the IDEFO process model [50,51] of the proposed crowd-based safety-rule violation labeling. The inputs of this process consist of three parts: (1) safety rules for properly executing certain construction activities; (2) workplace images that may contain cases that violate safety rules; and (3) the safety-related cognitive capability and “common sense” of online annotators, who are the participants of construction-site image labeling games. The authors made the following assumptions about the annotators’ cognitive capability and common-sense necessary for properly using the crowdsourcing system:

- The annotators could recognize the objects in images that capture workplace scenes if they knew the shapes of the objects.
- The annotators could correctly understand the safety-relevant spatial relationship between objects by looking at the images (e.g., object A is under object B).
- The annotators have common sense about how people could be hurt by objects or risky areas related to different energy sources (e.g., people fall from high elevation will be injured or killed; people touching objects with the electric current will be injured or killed).

The outputs of the crowdsourcing process are the probabilities of a list of predefined safety rules violated in the scenes captured in images, and the probabilities of having no safety-rule violations in the images assessed. The constraint is the crowd consensus model that will be discussed in detail in Section 3.3.

The mechanisms of this methodology are as follows. Workplace images that may contain violations will be uploaded to the online crowdsourcing platform. The image annotators will then play a violation annotation game to identify the violations of safety rules based on the workspace images. The game consists of three steps. (1) Training and quizzing—the annotators are divided into groups and asked to learn about certain safety rules using the training materials. After the training, the annotators take a quiz containing ten images about the safety rule they learned. The final violation labeling process will only use the annotation results from the annotators who achieve 70% or higher correctness in the quiz. (2) Annotating—the annotators who pass the quiz will annotate images violating certain safety rules that the annotators learned in the training process. (3) Violation labeling through annotation aggregation—this step will aggregate the answers from the annotators and then use the Bayesian-network-based crowd consensus model to determine whether a workplace image contains safety-rule violations.

The first subsection below uses a motivation experiment to define the two-types of majority vote mistakes that causing a high false-positive rate in safety-rule violation labeling. The second subsection then introduces how the training process resolves type-1 mistakes. The third subsection introduces how the Bayesian-network-based crowd consensus approach resolves type-2 mistakes.

### 3.1. Problem definition: two types of majority-vote mistakes made while using crowdsourcing

To understand the technical feasibility of utilizing crowdsourcing techniques for labeling images with safety-rule violations, the researchers conducted an experiment to test how online image annotators respond to the elevator-installation workplace images that might contain safety-rule violations. Fig. 2 shows the process of this experiment of crowdsourced safety-risk labeling. The input of this experiment involves three parts:

- 1) *Twelve safety rules* (shown in Table 1). The elevator company has 91 rules for the safety inspection of elevator installation scenes. In this experiment, the authors selected twelve rules that are violated the most frequently, according to historical data of the company.

- 2) *Twenty workplace images that may contain one safety violations* (shown in Fig. 3). Professional safety engineers determined the ground truths of the images’ safety-rule violations (the correct labels of images about the contained violations), as shown in Table 2.
- 3) *Two groups of online image annotators* (group  $\alpha$  and group  $\beta$ , 60 annotators in each group) recruited from the online crowdsourcing platform Amazon Mechanical Turk [52]. All annotators were allowed to participate in the image labeling tasks in both groups. These annotators were neither expected to have experiences in construction safety inspection nor to be trained about the detailed explanations of the safety rules used during the experiments.

To ensure the representativeness of this study, the hazards shown in the twenty workplace images covered all performance shaping factors (PSFs) addressed in the cognitive reliability and error analysis method (CREAM) [53]. CREAM is a mature and accurate version of risk behavioral studies [54]. The PSFs include human-related errors (e.g., observations, explanations), technology-related issues (e.g., equipment failures), and organization-related factors (training issues) [55], which provides a thorough framework for analyzing risks. Table 3 shows the detailed classification information of the PSFs and relationships between the PSFs and the images containing violations in Fig. 3. For example, the slash marks in column A represent the PSFs involved in image A in Fig. 3, which are human planning, organization and conditions in organization-related factors.

As shown in Table 3, the images containing violations in Fig. 3 involve all the performance shaping factors, which support the notion that the findings will be practical for the safety inspection. To further ensure the practicability of the experiment, the authors compared the distribution of violation and safety scenes in the twenty workplace images with the real world distribution data in Table 1. The K-S test showed that the p-value is 0.028, which cannot reject the hypothesis that the violation distribution in the experiments corresponded with the counterpart in the real practice of safety inspection records of the elevator company.

The researchers divided the twelve safety rules into two sets (safety rules 1-6 and safety rules 7-12). Each of the two groups of annotators was asked to choose the violated safety rule from one of the two sets of safety rules, respectively (i.e., group  $\alpha$  annotated rules 1-6 and group  $\beta$  annotated rules 7-12). The authors used the choice of the most image annotators as the safety-rule violation labeling results. Such labeling results were the “majority votes.” Then the authors integrated the answers of the two groups to conclude the safety-rule violation labeling result of each image. Table 2 lists those images, along with the violated rules and violation labeling results. The results showed that the majority votes misclassified nine of eleven images that contain no violations as “containing a violation.” Majority votes also incorrectly labeled six of nine images with violation rules different from the actual rules violated in the scenarios. The false-positive rate is up to 75%, which would require extra attention from professional safety inspectors.

The researchers then considered the violation labeling results for image-rule pairs to identify the reasons for the false positives produced by majority votes. The authors categorized the image-rule pairs into three types as shown in Table 4:

- Image-rule pair type 1: The image contains a violation, while the violated safety rule is among the choices for annotators’ selections.
- Image-rule pair type 2: The image contains a violation, but the violated safety rule is not among the choices for annotators’ selections.
- Image-rule pair type 3: The image contains no safety violation.

Fig. 4 shows an example of the three types of possible image labeling results by majority votes. The correct label is represented in the green color with the highest percentage in Fig. 4(a), (b) and (c). Fig. 4(a) shows the result of the image T labeling results in group  $\alpha$ , as

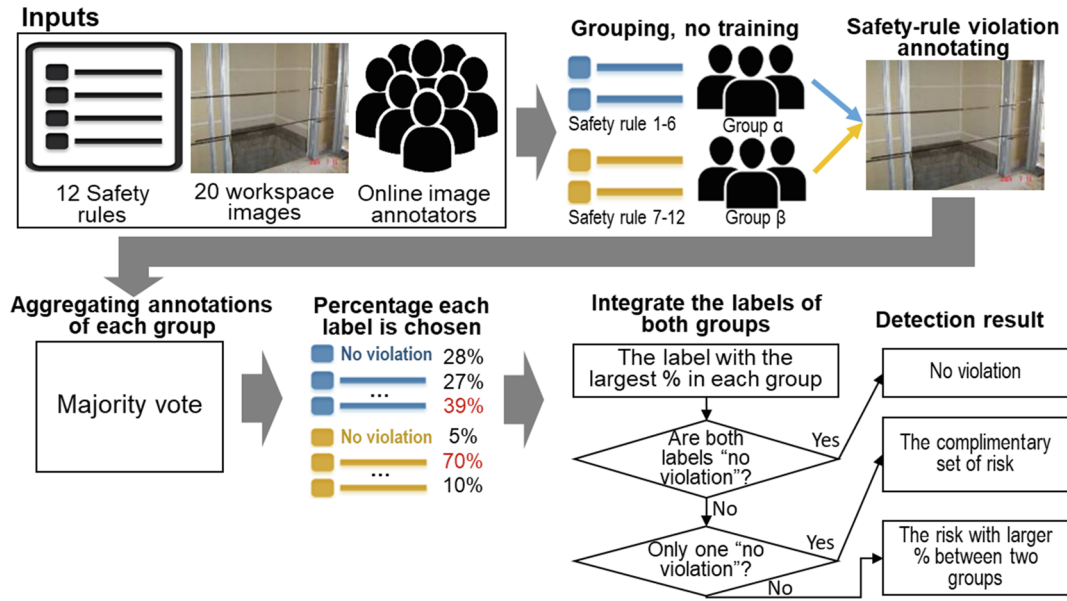


Fig. 2. The experiment process of crowdsourced safety-violation labeling.

an example of the category “correct labeling result.” In this case, the majority-vote answer was correct: the answer matched the violated safety rule. However, such correct label in all results was limited. As shown in Table 5, the false label results accounted for 75% (fifteen out of twenty). Such results indicated the capability of non-experts to identify workplace hazards based on their “common sense.” The crowdsourcing method was viable due to such use of common sense. The authors further analyzed those incorrect result and identified two kinds of Majority-vote mistakes, as follows:

- Majority-vote mistake type 1 (Fig. 4b): More than 50% of the annotators chose the same wrong answer. Such agreement on a wrong answer indicated that the crowd has a strongly biased understanding of a certain safety rule.
- Majority-vote mistake type 2 (Fig. 4c): The opinions of the annotators are diverse. The highest percentage of image annotators that chose the same answer is less than 50%. Such diverse answers reflected the confusion and disagreements among people about the image contents. In such cases, slight biases of image annotators could cause a switch of the “majority-vote answer.”

Fig. 4(b) shows the majority vote mistake type 1, which is the annotation result of image A in group α, more than 53% of the annotators chose the same wrong answer, as represented in the dark brown color. Fig. 4(c) shows the majority vote mistake type 2, which is the annotation result of image B in group β. The answer chose by most people

was wrong, but the percentage gaps between each answer were smaller than that in Fig. 4(b). Mistake types 1 and 2 exist in the results for all types of image-rule pairs, as summarized in Table 5. The new crowdsourcing methods aim at addressing these two types of mistakes so as to overcome the false positives generated by majority votes.

### 3.2. The training step for resolving type-1 mistakes of majority votes

The purpose of the training step was to help the crowdsourcing annotators who had no prior knowledge about workplace safety understand the safety rules in less than ten minutes. The training process needed to be informative and easy to understand for people who had no experience in construction safety inspection. There were two steps in the training process of the crowdsourcing platform:

- 1) Training. The training process provided the image annotators six safety rules so that the cognitive workload of the annotators would be suitable for short training of ordinary online annotators, compared with annotating the 91 safety rules of the elevator installation, which required years of professional training in safety inspection.
- 2) Quizzing. The goal of the quiz was to filter out the annotators who learn too little from the training, based on their quiz scores. Such filtering could help delete unreliable labels contributed by under-performers. The quiz consisted of ten images that may contain violations (Fig. 5).

Table 1

Twelve rules that have the highest Disqualification Percentage (DP) in elevator installations.

| Rule No. | Description                                                                                   | DP    |
|----------|-----------------------------------------------------------------------------------------------|-------|
| 1        | Improper working platforms on scaffold structure                                              | 6.26% |
| 2        | Improper door blocking device                                                                 | 4.84% |
| 3        | Failure to provide overhead protection while working in the hoistway                          | 3.88% |
| 4        | Failure to use ground fault circuit interrupters or equivalent protective devices             | 4.07% |
| 5        | Inadequate electrical protection in the proximity of work activity                            | 2.67% |
| 6        | Slings not protected against sharp edges                                                      | 1.50% |
| 7        | Improper guardrail system                                                                     | 1.48% |
| 8        | Working in proximity to unguarded drive or diverter sheaves or other rotating equipment       | 1.19% |
| 9        | Employees working in a wet pit with power on                                                  | 0.91% |
| 10       | Jewelry and other metallic objects were worn around live equipment                            | 0.48% |
| 11       | Unsafe oxygen-acetylene or compressed gases welding, cutting, heating equipment or procedures | 0.31% |
| 12       | Storage of workplace materials creating an unsafe mechanical energy source                    | 3.56% |

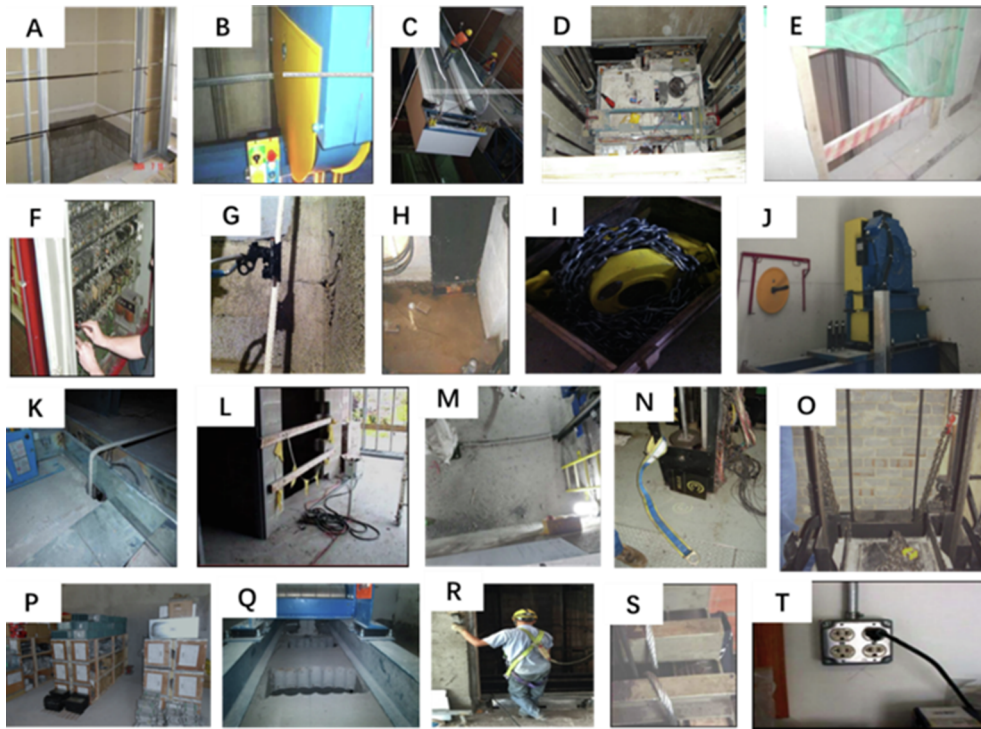


Fig. 3. Twenty workplace images that may contain violations of safety rules.

Table 2

Safety-violation labeling result in the preliminary experiments.

|                             |                   |                  |                   |                  |                  |
|-----------------------------|-------------------|------------------|-------------------|------------------|------------------|
| Workplace image No.         | A                 | B                | C                 | D                | E                |
| Ground truths of violations | Violating Rule 7  | No violation     | No violation      | No violation     | Violating Rule 7 |
| Labeling results            | Violating Rule 2  | Violating Rule 8 | Violating Rule 1  | Violating Rule 8 | Violating Rule 2 |
| Workplace image No.         | F                 | G                | H                 | I                | J                |
| Ground truths of violations | Violating Rule 10 | No violation     | Violating Rule 9  | No violation     | No violation     |
| Labeling results            | Violating Rule 9  | Violating Rule 1 | Violating Rule 8  | No violation     | Violating Rule 6 |
| Workplace image No.         | K                 | L                | M                 | N                | O                |
| Ground truths of violations | Violating Rule 6  | Violating Rule 7 | No violation      | No violation     | No violation     |
| Labeling results            | Violating Rule 6  | Violating Rule 2 | Violating Rule 1  | Violating Rule 5 | Violating Rule 8 |
| Workplace image No.         | P                 | Q                | R                 | S                | T                |
| Ground truths of violations | No violation      | No violation     | Violating Rule 10 | Violating Rule 6 | Violating Rule 4 |
| Labeling results            | No violation      | Violating Rule 8 | Violating Rule 9  | Violating Rule 6 | Violating Rule 4 |

Table 3

The relation between the PSFs with the images containing violation in Fig. 3.

|                             |                            | Workplace image No. |   |   |   |   |   |   |   |   |  |
|-----------------------------|----------------------------|---------------------|---|---|---|---|---|---|---|---|--|
| Performance Shaping Factors |                            | A                   | E | F | H | K | L | R | S | T |  |
| Human                       | Observation                |                     |   | / | / |   |   | / |   |   |  |
|                             | Explanation                |                     |   | / | / |   |   | / |   | / |  |
|                             | Planning                   | /                   |   |   |   |   | / |   |   |   |  |
|                             | Temporary function related |                     |   |   |   | / |   |   |   | / |  |
|                             | Permanent function related |                     | / |   |   |   |   |   |   |   |  |
| Technology                  | Equipment                  |                     |   |   |   |   | / | / |   |   |  |
|                             | Temporary interface issues |                     |   |   |   |   |   |   |   | / |  |
| Organization                | Organization               | /                   | / |   |   |   | / |   |   |   |  |
|                             | Training                   |                     |   | / | / |   |   | / |   |   |  |
|                             | Surrounding                |                     |   | / |   |   |   |   |   |   |  |
|                             | Conditions                 | /                   | / |   |   |   | / |   |   |   |  |

In the first step, the training materials for teaching each safety rule showed the images of the violation scenes, along with compliance scenes and word descriptions. Images came from one elevator installation company. From the Elevator Industry Field Employees' Safety Handbook [56], the authors summarized some explanations of the rules. The images together with word explanations helped the image annotators learn to detect hazards from job site images. For each rule, the authors tried to use the minimum images and words. Specifically, in one rule, the average training images number was four, and the explanations were limited in two sentences.

The second step, quizzing, was to ensure that the annotators comprehended correct knowledge from our training. The training effect could vary from people to people, due to their understanding and thinking. The authors prepared ten images as the quiz to test the comprehension levels of the annotators. As shown in Fig. 5, the quiz images were different from the training materials and the violation labeling materials. After the training, annotators who gained 70% correct rates in the quiz became trustable image annotators. In the quiz,

**Table 4**  
Three types of image-rule pairs.

| Image-rule pair types | Image-rule pair type 1                                                                  | Image-rule pair type 2                                                                   | Image-rule pair type 3                                                                    |
|-----------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Example               | <br>12 | <br>1-6 | <br>12 |

image IX, VI, and VIII violate rules 1, 5 and 6 respectively, and image VI, II, and I violate rule 8, 10 and 12 respectively. The other four images contain no violation.

As an example of the training process, Fig. 6 shows the training material for the safety rule “Improper door blocking device” (upper half of the figure) and a testing image along with this image’s labeling results generated by annotators who labeled the images with or without the training (lower half of the figure). In the experiment where the annotators did not take the training, 53% of the annotators chose “Improper door blocking device” as the violation in the image due to the lack of understanding about what that rule means, while the correct answer should be “No violation.” In this example, the majority of annotators chose the same wrong answer, which is a type-1 mistake. However, in the experiment including the training process, the percentage of annotators that mistakenly choose “Improper door blocking device” was reduced to 33% because the training process clarified this safety rule. The annotators realized that the scenario in the workplace image, as shown in the lower half of Fig. 6, was not relevant to door blocking devices. The training process, therefore, transformed the type-1 majority-vote mistake into type-2.

Table 6 summarizes the type-1 majority-vote mistakes in experiments with and without the training processes. The authors observed no type-1 mistakes in the experiments with training, while there were equal or more than a quarter type-1 majority-vote mistakes in the experiment without training. Such results indicated that the training process could effectively correct the biased understanding of the safety rules by the online image annotators. With the help of the training session, the high concentration of wrong answers became wrong answers with higher diversities (i.e., type-1 majority vote mistakes become type-2 majority-vote mistakes). The crowd consensus model presented below could handle type-2 mistakes correctly and annotate the images as “no violation.”

### 3.3. A Bayesian-network-based crowd consensus model for resolving the type-2 mistakes of the majority votes

The previous section shows that the training step could effectively convert the majority-vote mistakes in crowdsourcing violation labeling from type-1 to type-2. The type-2 majority-vote mistakes were distinguishable from a true violation because in type-2 mistakes each wrong answer was occupying a small portion of all responses, while the answer of a true violation would occupy the majority of the responses. With the training process, the labelers were highly likely to choose the violated safety rule if the image contained a safety violation and the violated safety rule was among the choices. On the other hand, when the image contained no safety rule violation, or the violated safety rule was not among the choices, the image annotators were less likely to choose the correct answer of “no violation,” but may diversely choose a non-violated safety rule from the choices. This fact motivated the development of a Bayesian-network-based crowd consensus model. A crowd consensus model uses a probabilistic approach to aggregate the evaluations from the set of evaluators for overcoming the noises introduced by confusions of annotators in annotating images having no violations, as detailed below. Specifically, the crowd consensus model aims to determine the probability of the violation of a rule in an image, given responses from the group who has learned this rule.

The rationale behind selecting the Bayesian-network-based crowd consensus model is that the Bayesian-network model is a generative model that reflects the pattern of image interpretation mentioned above. When one of the safety rules is selected as violated by the largest portion of image annotators, the crowd consensus model will assess the probability of each answer being correct. If the percentage of annotators choosing the most selected safety rule is not high enough, the crowd consensus model will assess this image as “no violation” because the probability of “no violation” being correct will be the highest, although it is not the answer that selected by most people. On the other hand, if “no violation” is the most-selected answer, then the correct

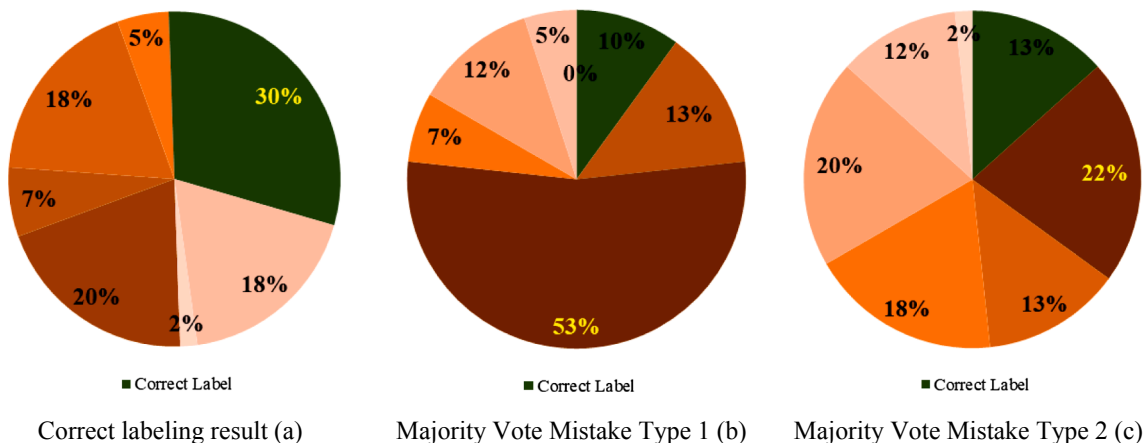


Fig. 4. Three types of labeling results of majority vote.

**Table 5**

Summary of the majority vote mistake type.

| Image-rule pair types                           | Image-rule pair type 1   |        | Image-rule pair type 2 |        | Image-rule pair type 3 |
|-------------------------------------------------|--------------------------|--------|------------------------|--------|------------------------|
| Correct Label                                   | The violated safety rule |        | No Violation           |        |                        |
| False labeled images number/Total images number | 6/9                      |        | 9/11                   |        |                        |
| Majority Vote Mistake Type                      | Type 1                   | Type 2 | Type 1                 | Type 2 | Type 2                 |
| Majority Vote Mistake Type Rate                 | 67%                      | 33%    | 37%                    |        | 63%                    |

label of this image-rule pair is “no violation.” The assumptions of the Bayesian network model are as follows:

- Annotators are independent, i.e., the annotation of one workplace image from one annotator does not affect other annotators.
- The probability of each rule violated in a given image is independent of other images.
- Each workplace image may contain at most one violation of the given safety rules.
- The probabilities of annotators correctly identifying the violations of different safety rules in images with simple and diverse backgrounds are roughly the same. In future studies, the researchers will explore how various safety rules and workplace images with different visual complexity will influence the annotators' ability to identify the violations.
- When an annotator chooses a wrong answer, the probabilities of choosing different wrong choices are the same [16].

Fig. 7 shows a graphical representation of this crowd consensus model. The crowdsourcing process asks  $N$  annotators (each annotator is assigned a number  $n$ ) to label  $I$  images (each image is assigned a number  $i$ ) against  $K$  rules (each rule is assigned a number  $k$ ). The input of the crowd consensus model are the responses about image  $i$  from  $N$  annotators:

$$A_i = \{a_{i0}, a_{i1}, \dots, a_{iK}\}, \sum_{k=0}^K a_{ik} = N \quad (1)$$

where  $a_{ik}$  is the number of annotators that choose rule  $k$  ( $k \in [0, K]$ ) for image  $i$  ( $i \in [1, I]$ ). Notice that  $k = 0$  refers to the answer “no violation.” Therefore,  $a_{i0}$  denotes the number of annotators who think that image  $i$  contains no violation against the rules they have learned in the training step, while  $a_{ik}$  ( $k \neq 0$ ) denotes the number of annotators who think the image  $i$  contains violations against safety rule  $k$ .

The authors define the ground truth of image  $i$  capturing a violation of safety rule  $k$  or no violation as  $\Phi_i = 0, 1, 2, \dots, K$ .  $\Phi_i = 0$  means that image  $i$  contains no violation. The authors also define  $r_{ikn} \in \{0, 1\}$  as the annotation result of image  $i$  with respect to rule  $k$  by the annotator  $n$  ( $n \in [1, N]$ ), while the integrated labeling result of image  $i$  by all

annotators is  $\Phi_i^*$ .  $p_{ikn}$  is the probability of annotator  $n$  correctly choosing the answer  $k$  for image  $i$ , which is the function of annotator  $n$ 's cognitive ability, the difficulty of understanding safety rule  $k$ , and the visual complexity of image  $i$ . The expected result from the violation annotation is the probability of the event “the image contains a violation of rule  $k$ ” when  $M$  out of  $N$  annotators chose rule  $k$  as the violation captured in image  $i$ :

$$P(\Phi_i = k | a_{ik} = M) \quad (2)$$

Based on the Bayes rule:

$$P(\Phi_i = k | a_{ik} = M) = P(a_{ik} = M | \Phi_i = k) \cdot P(\Phi_i = k) / P(a_{ik} = M) \quad (3)$$

Each item on the right-hand side of Eq. (3) has a practical meaning. The following bullets will introduce the calculation of these right-hand side items in detail.

- Calculating  $P(a_{ik} = M | \Phi_i = k)$ :

$P(a_{ik} = M | \Phi_i = k)$  is the probability of having  $M$  out of  $N$  image annotators choose rule  $k$  when  $k$  is the correct answer. Calculating  $P(a_{ik} = M | \Phi_i = k)$  requires knowing the probability of having  $M$  out of  $N$  people choose rule  $k$  as the correct answer when the ground truth  $\Phi_i = k'$ ,  $k' = 0, 1, \dots, K$ :

$$P(a_{ik} = M | \Phi_i = k) = \binom{N}{M} p_{ikn}^M (1 - p_{ikn})^{N-M} \quad (4)$$

where  $p_{ikn}$  is the probability that the image annotator  $n$  chooses answer  $k$  as the label of image  $i$ :

$$p_{ikn} = \begin{cases} p_{CVI}, & \text{when } k = k' \neq 0 \\ p_{TN}, & \text{when } k = k' = 0 \end{cases} \quad (5)$$

In this study, the authors assume that  $p_{ikn}$  contains two values  $p_{CVI}$  and  $p_{TN}$  for correctly annotating images with/without a violation regardless of different qualified annotators and safety rules. Section 4.2 will validate that the safety-rule labeling accuracy of the developed crowd consensus model is robust with this assumption, even when the value of  $p_{CVI}$  and  $p_{TN}$  vary in certain ranges.  $p_{CVI}$  is the prior correct violation identification rate of a response  $r_{ikn}$ , which is the prior knowledge of the average probability of image annotators correctly

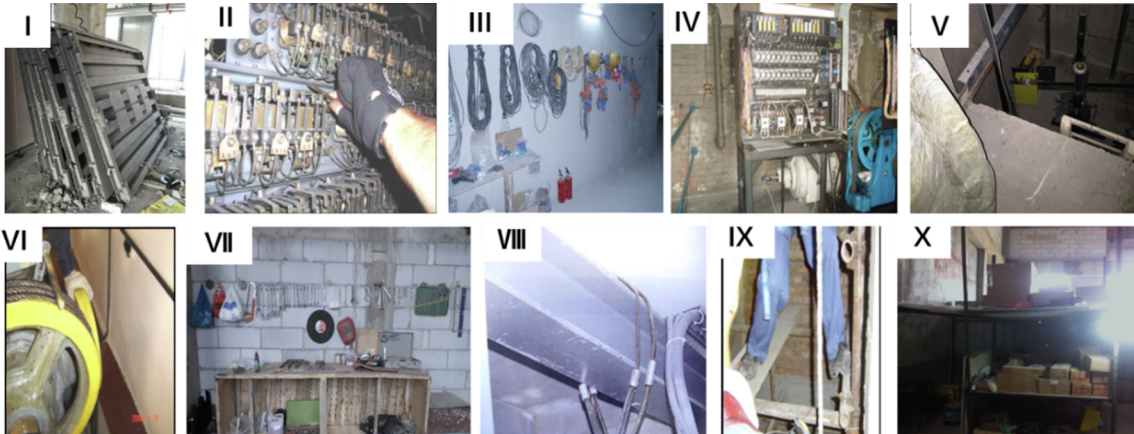


Fig. 5. Ten workplace images that may contain violations of safety rules for the quiz.

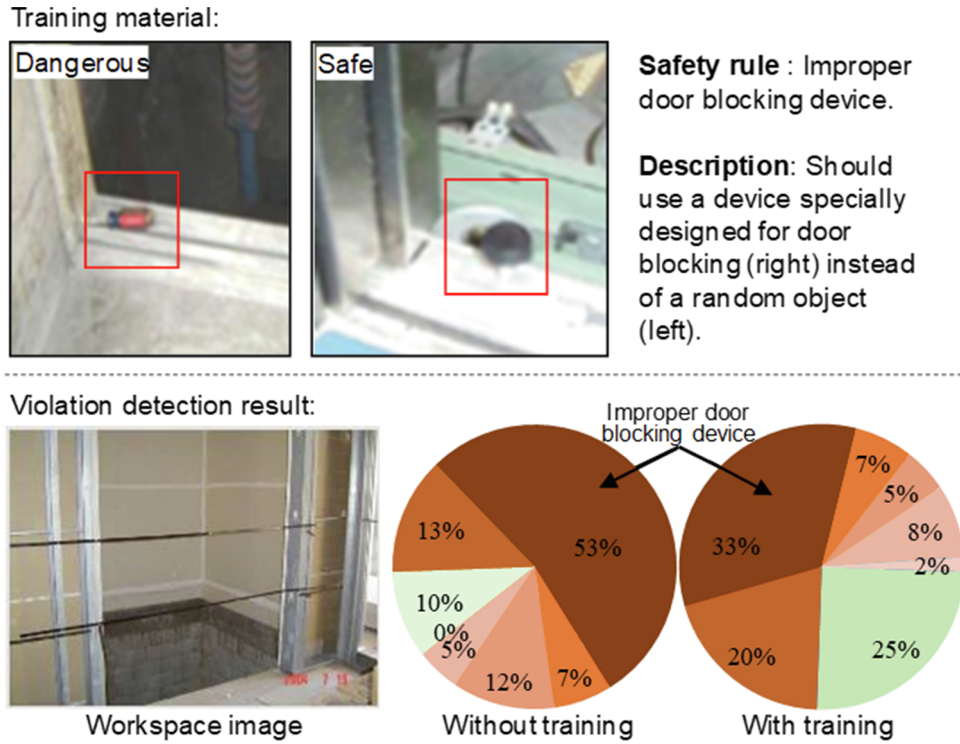


Fig. 6. The training process: the training material and its effect.

**Table 6**  
Summary of type-1 majority vote mistake.

| Type-1 Majority Vote Mistake Rate | Groups           | Group $\alpha$ | Group $\beta$ |
|-----------------------------------|------------------|----------------|---------------|
|                                   | Without Training | 0.25           | 0.30          |
|                                   | With Training    | 0.00           | 0.00          |

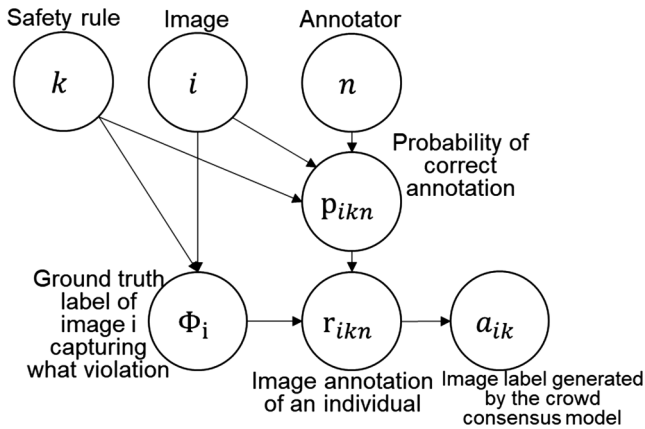


Fig. 7. Graphical representation of the Bayesian-network-based crowd consensus model.

selecting the violated safety rule of a workplace image when the violated safety rule is among the choices. Similarly,  $p_{TN}$  is the prior true negative rate, which is the prior knowledge of the average probability of image annotators correctly selecting “no violation” when the workplace image contains no violation, or the violated safety rule is not among the choices.

$p_{CVI}$  and  $p_{TN}$  can be calculated based on the results of online image labeling games where the ground-truth violation label of each workplace image is known, which is the preliminary training data of the crowd consensus model. In the result,  $M_1$  out of  $N_1$  annotators chose the

correct violated rule in all images that contain violations. To estimate the value of  $p_{CVI}$  using maximum likelihood estimation, the authors generated the likelihood function:

$$L(p_{CVI}, M_1) = P(a_{ik}=M|p_{CVI}) = \binom{N_1}{M_1} p_{CVI}^{M_1} (1 - p_{CVI})^{N_1-M_1} \quad (6)$$

To derive  $p_{CVI}$ , the authors maximized the likelihood function by differentiating (6) with respect to  $p_{CVI}$  and setting to zero:

$$p_{CVI} = \underset{p_{CVI}}{\operatorname{argmax}} L(p_{CVI}, M_1) = \frac{M_1}{N_1} \quad (7)$$

The authors calculate  $p_{TN}$  in a similar manner. In the training data, suppose that  $M_2$  out of  $N_2$  annotators chose “no violation” correctly for all images that contain no violations. The authors derive  $p_{TN}$  by maximizing the likelihood function:

$$p_{TN} = \underset{p_{TN}}{\operatorname{argmax}} \left( \binom{N_2}{M_2} p_{TN}^{M_2} (1 - p_{TN})^{N_2-M_2} \right) = \frac{M_2}{N_2} \quad (8)$$

- Acquiring  $P(\Phi_i = k)$ :

$P(\Phi_i = k)$  is the probability that rule  $k$  is violated in a workplace image which is provided by the historical data about the chances of having violations appear in random field observations or randomly-taken field images. This probability  $P(\Phi_i = k)$  is a site- and business-specific domain knowledge for calibrating the crowd consensus model. Such knowledge should be derived based on a sample of workplace images selected by experienced civil engineers. In this study,  $P(\Phi_i = k)$  is the safety rule violation distribution of the images in Table 2. These images are selected by the safety engineers from the elevator installation company as representative samples based on their working experiences in many elevator installation work spaces under the regulation of the company's safety standards.

- Calculating  $P(a_{ik} = M)$ :

$P(a_{ik} = M)$  is the expectation of the probability that  $M$  number of  $N$  annotators chose the answer “ $k$ ”, where “ $k$ ” represents a choice of the image annotating problem selected by an annotator, which can be calculated by:

$$P(a_{ik} = M) = \sum_{k'=0}^K P(a_{ik}=M|\Phi_i = k') * P(\Phi_i = k') \quad (9)$$

where

$$P(a_{ik}=M|\Phi_i = k') = \binom{N}{M} p_{ikn}^M (1 - p_{ikn})^{N-M}, \quad (10)$$

$$p_{ikn} = \begin{cases} p_{CVI}, & \text{when } k = k' \& k' \neq 0 \\ \frac{1-p_{CVI}}{K}, & \text{when } k \neq k' \& k' \neq 0 \\ p_{TN}, & \text{when } k = k' \& k' = 0 \\ \frac{1-p_{TN}}{K}, & \text{when } k \neq k' \& k' = 0 \end{cases} \quad (11)$$

In Eq. (9), the “ $k$ ” can be any number from 0 to 6 in our experiment, wherein  $k = 0$  means no safety rule is violated in the workplace image. Because the training session will influence the probability of the annotators selecting each answer, the authors calculated the average probability that  $M$  number of  $N$  annotators chose the answer “ $k$ ” when the correct answer is any  $k' = 0, 1, 2, \dots, K$ . Therefore, the authors calculated the conditional probability that  $M$  number of  $N$  annotators chose the answer “ $k$ ” considering different “ $k$ ” and then get the expectation by considering  $P(\Phi_i = k')$ .

Finally, the labeling result of image  $i$ , denoted as  $\Phi_i^*$ , is the choice  $k$  that has the highest probability to be correct:

$$\Phi_i^* = \underset{k}{\operatorname{argmax}} (P(\Phi_i = k | a_{ik} = M)) \quad (12)$$

#### 4. Validation

The authors selected twelve safety rules with the highest disqualification percentage (shown in Table 1) and collected 66 elevator installation workplace images for the validation experiment. The authors chose twenty images, including nine images containing different violations and some safety-rule benign images (shown in Fig. 3), as testing images to assess the labeling performance of the crowd consensus model with various inputs. The authors also chose ten different images to set up a quiz to test the impacts of training on the performance of the annotators (shown in Fig. 5). In the quiz, image IX, VI, and VIII violate rules 1, 5 and 6 respectively, and image VI, II, and I violate rules 8, 10 and 12 respectively. The other four images contained no violation. The authors then used the remaining 36 images to create the training material for the twelve safety rules, providing both violation and benign examples for each safety rule. The two groups of online image annotators (group  $\alpha_1$  and  $\beta_1$ ) recruited from Amazon Mechanical Turk went through the training process to learn the basic knowledge about six safety rules (group  $\alpha_1$  learned rule 1-6 and group  $\beta_1$  learned rule 7-12).

All image annotators needed to take a quiz before annotating 20 images based on the safety rules learned by them. Table 7 shows the correct rates (i.e.,  $p_{CVI}$  and  $p_{TN}$  for images with or without safety violations, respectively) in the annotation results of the annotators. The table separates the correct rates of the annotators who passed the quiz or not. The T-test result, with the p-value at 0.003, validates the

significant difference between the answers from the annotators who passed the quiz and those contributed by those who failed the quiz. With the p-value at 3.46E-05, the T-test also validates the significant difference between the  $p_{CVI}$  and  $p_{TN}$  for the annotators who passed the quiz, while no significant difference exists for the annotators who failed the quiz. These results indicate that 1) the quiz can identify the annotators with high capability and reliability to improve the annotation performance, and 2) the assumption that annotators are more sensitive to violations rather than no-violation images is valid.

Finally, the crowd consensus model integrated all answers from both group's annotators (group  $\alpha_1$  and  $\beta_1$ ) who passed the quiz to calculate the probabilities of the safety rules violated in each image, as well as the probability of having no violation in each image. The algorithm then determined the final violation label of the workplace images. The authors compared the labeling results generated through the crowd consensus model against the results generated through majority votes. This section first verified the performance of the training step for addressing the type-1 majority vote mistakes, and then verified the performance of the Bayesian-network-based crowd consensus model for addressing the type-2 majority vote mistakes.

The following subsections will examine how different parameters influence the performance of the proposed crowd consensus model for addressing type-2 mistakes. The performance indices of the proposed crowd consensus model are the true negative rate and the correct identification rate:

- POsterior True Negative Rate ( $po_{TN}$ ): the number of images that are correctly interpreted as containing no violation, divided by the number of images containing no violation of any safety rule in the experiment result.
- POsterior Correct Violation Identification Rate ( $po_{CVI}$ ): the number of images that are correctly interpreted as violating a certain safety rule, divided by the number of images containing a violation of any safety rule in the experiment result.

Two factors of this crowdsourcing method can influence the performance of safety-rule violation labeling. The first factor is the crowd size. The first subsection will discuss how the number of annotators influences the performance of crowdsourcing-based violation labeling in construction projects. For any given crowd size from 10 to 50, the authors randomly subsampled the annotators' answers to that size for 1,000 times.

The second factor is the parameters in the crowd consensus model: the prior correct identification rate  $P_{CVI}$  and the prior true negative rate  $P_{TN}$ . The default value of  $P_{CVI} = 0.7152$  and  $P_{TN} = 0.3517$  is calculated based on the result of previous image interpretation games. The dataset for automatic workplace safety monitoring and diagnosis needs both the safety-rule violation and compliance workplace image datasets. For filter out false alarms, the data set needs the crowd consensus model to have a near 100%  $po_{TN}$ . On the other hand, for efficiently acquiring workplace images violating certain safety rules, the dataset needs the crowd consensus model to have a relatively high  $po_{CVI}$ . The second subsection will introduce how the changing of  $P_{CVI}$  and  $P_{TN}$  influences the performance of indicators of the violation labeling result,  $po_{CVI}$  and  $po_{TN}$ .

##### 4.1. Evaluating the violation labeling performance concerning different crowd sizes

Fig. 8 compares the performance of the crowd consensus model and “majority vote” in crowdsourcing violation labeling. Fig. 8(a) compares the  $po_{TN}$  of the labeling result of majority vote and crowd consensus model for workplace images with no violation. In Fig. 8(a), the crowd consensus model steadily achieved 100%  $po_{TN}$  when the crowd size was larger than 30, while the majority vote could only achieve less than 40%  $po_{TN}$ . Also, the posterior true negative rate of the result using the

**Table 7**

The correct rate ( $p_{CVI}$  or  $p_{TN}$ ) of annotators who passed or failed the quiz.

| Average Correct Rate       | $P_{TN}$ | $P_{CVI}$ |
|----------------------------|----------|-----------|
| Annotators passed the quiz | 0.418    | 0.783     |
| Annotators failed the quiz | 0.236    | 0.210     |

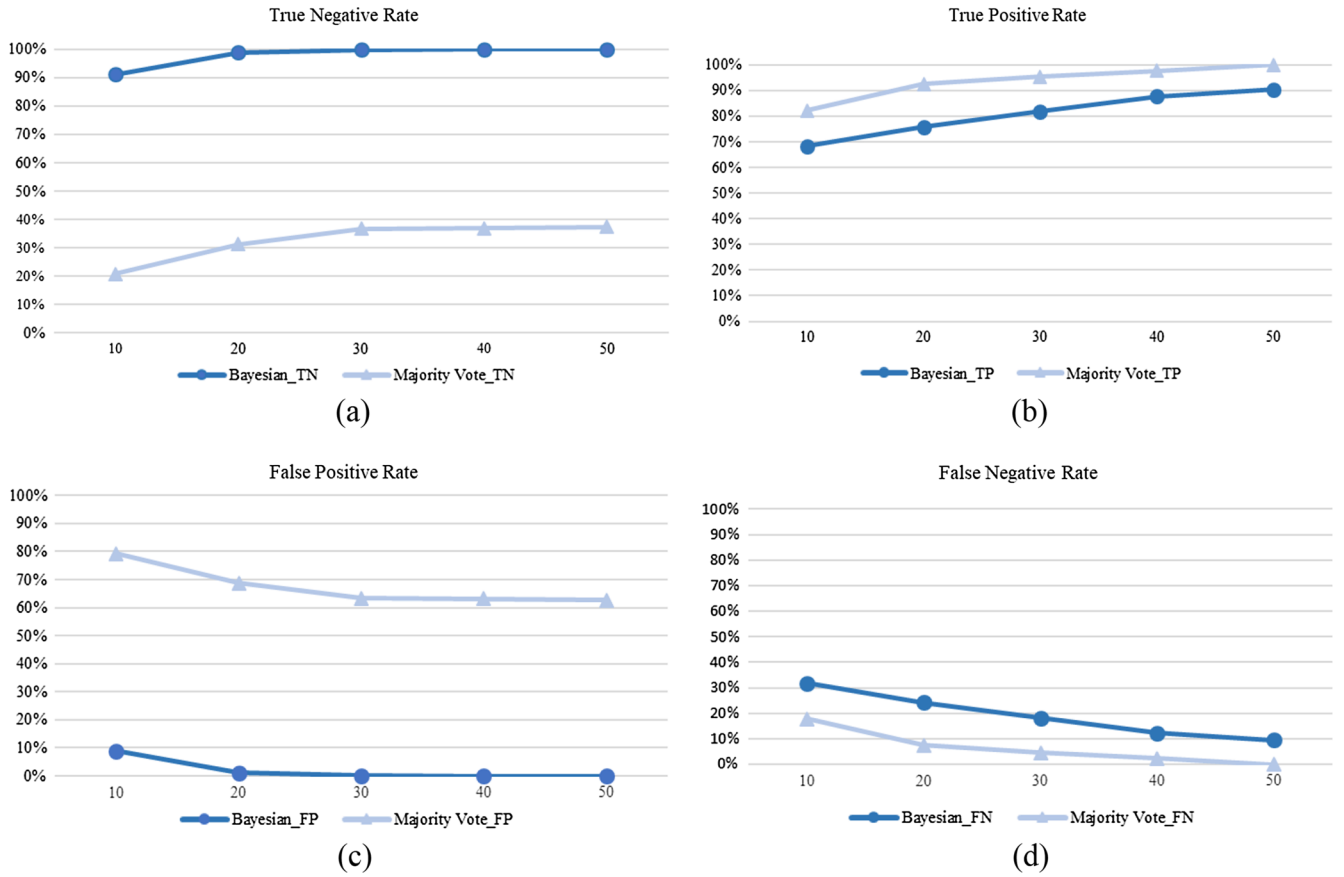


Fig. 8. Comparing the performance of the crowd consensus model and “majority vote” in crowdsourcing violation labeling: (a)  $po_{TN}$ , (b)  $po_{CVI}$ , (c)  $po_{FP}$  and (d)  $po_{FN}$  of the experiment result concerning different crowd size (x-axis).

crowd consensus model was higher than 90%, even at the crowd size of 10 people, while the counterpart using majority vote was about 20%. This result showed that the crowd consensus model could effectively improve the true-negative rate of workplace risk analysis against majority votes with as few as ten annotators.

Fig. 8(b) compares the  $po_{CVI}$  of the violation labeling results of majority vote and crowd consensus model for workplace images with a violation. The  $po_{CVI}$  was higher than 80% of both crowd consensus model and the majority votes when the crowdsize was 50, which showed that the violation identified by the crowdsourcing could be a reliable reference for establishing the safety-rule violation datasets.

Fig. 8(c) and (d) compares the false positive and false negative rate of the violation labeling results of the majority vote and crowd consensus model. The false positive rate is equal to the value of one minus  $po_{TN}$ . The false negative rate is equal to the value of one minus  $po_{CVI}$ . In this figure, the false positive rate has been significantly reduced in our crowd consensus model, and the false negative rate could be reduced to a relatively low level. Since many existed AI algorithms could handle labels with false-negative noisy, the labelling result of crowd consensus model would be practical for training advanced computer vision algorithms.

Fig. 9 compares the correct rate of the majority vote and the crowd consensus model. The correct rate is the total images number, which is twenty in this experiment, divided by the number of images correctly labeled by the algorithm. The performance of the crowd consensus model could achieve 90% correctness when the crowd size was 30. The majority vote model correct rate also improved when the crowd size increased. However, the majority vote model could hardly achieve 70% correct rate, even if the crowd size was 50. Overall, the  $po_{TN}$  and  $po_{CVI}$  of the crowd consensus model could achieve 100% and 90% respectively with the crowd size of 50 people. The crowd consensus model

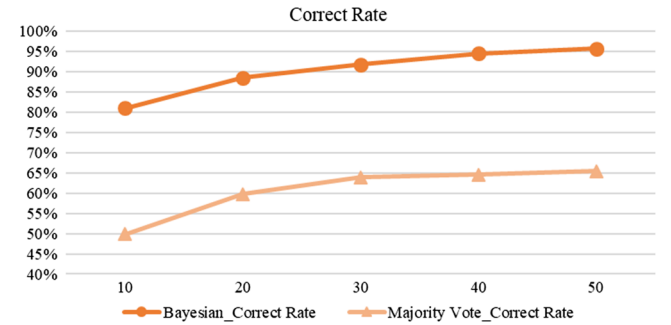


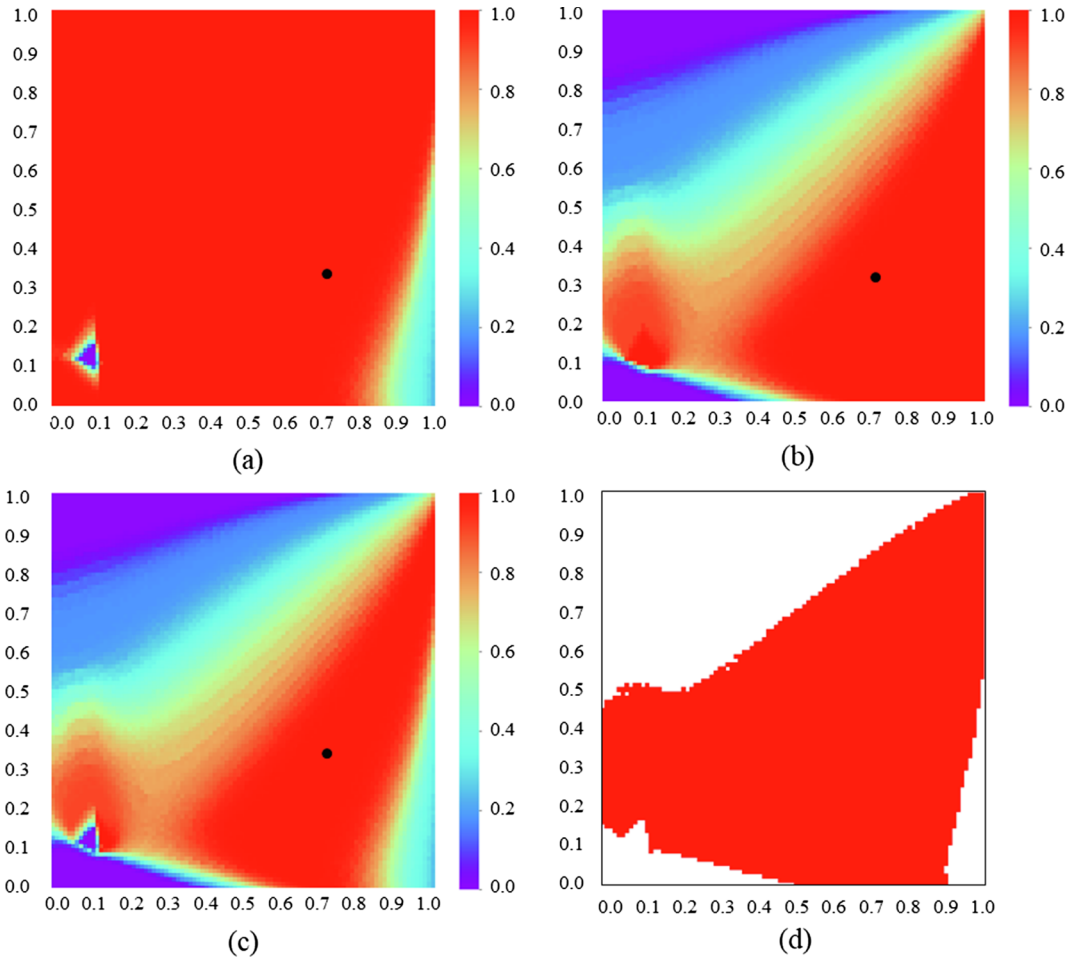
Fig. 9. Comparing the performance of the crowd consensus model and “majority vote” in crowdsourcing violation labeling.

could achieve 95% correct rate with the crowd size of 50 people.

#### 4.2. Evaluating violation labeling performance concerning different parameters ( $P_{CVI}$ and $P_{TN}$ ) in the crowd consensus model

Adjusting the parameter of  $P_{CVI}$  and  $P_{TN}$  can improve the  $po_{CVI}$  of violation labeling of the crowd consensus model with the cost of reducing the  $po_{TN}$ , and vice versa. Using the violation labeling result of 40-people crowd size as an example, Fig. 10 visualizes the impact of the adjustment of parameters of  $P_{CVI}$  and  $P_{TN}$  on the  $po_{TN}$ ,  $po_{CVI}$  and correct rate, as the following:

Fig. 10 (a)-(b) show the heat maps of the  $po_{TN}$  and  $po_{CVI}$  concerning the change of  $P_{CVI}$  (x-axis) and  $P_{TN}$  (y-axis), respectively. The black dots visualize the default parameter of  $P_{CVI} = 0.7152$  and  $P_{TN} = 0.3517$ , which will balance the performance of  $po_{TN}$  and  $po_{CVI}$ . To limit false alarms in the dataset and improve the efficiency of AI violation



**Fig. 10.** Violation labeling indexes concerning  $P_{CVI}$  (x-axis) and  $P_{TN}$  (y-axis) when crowd size = 40: (a)  $p_{CVI}$ ; (b)  $p_{TN}$ ; (c)  $\min(p_{CVI}, p_{TN})$ ; and (d) area that correct rate of the crowd consensus model outperform the counterpart of majority vote.

learning, the engineers could increase  $P_{TN}$  and/or reduce  $P_{CVI}$  near the neighborhood of point of coordinates (0.7152, 0.3517) and the performance of  $p_{CVI}$  would increase (Fig. 10(a)) while the  $p_{TN}$  could maintain at a high rate (Fig. 10(b)). For example, increasing  $P_{TN}$  and/or reducing  $P_{CVI}$  rate (say, set  $P_{CVI} = 0.66$ ;  $P_{TN} = 0.48$ ) could increase the  $p_{CVI}$  from 87% to around 92% while keeping the  $p_{TN}$  at 100%, which would be more practical for training of advanced computer vision algorithms.

Fig. 10(c) visualizes the heat map of the minimum of the true negative rate and the correct-risk-identification rate. In this figure, the point of (0.7152, 0.3517) is in the area with the highest “temperature.” This result showed that  $P_{tn}$  and  $P_{cvi}$  learned from the training data will balance the performance of true negative rate, and correct-risk-identification rate of the violation labels. Fig. 10(d) portrays a red area where the correct rate (the sum of true positive and true negative cases, divided by the total number of images) of the crowd consensus model outperformed the counterpart of the majority vote method. This figure shows that the performance of the crowd consensus model was robust and could surpass the majority vote, even when the parameter of  $P_{tn}$  and  $P_{cvi}$  deviated from their preliminary value.

The researchers further evaluated violation labeling performance concerning different crowd sizes and different parameters in the crowd consensus model. Fig. 11 further shows the tradeoff between the  $p_{TN}$  and  $p_{CVI}$  when  $P_{tn}$  and  $P_{cvi}$  changed for different crowd sizes.

The solid lines in Fig. 11 visualize the change of correct violation identification rate concerning the changing of the posterior false positive rate, which equals to the value of one minus  $p_{TN}$  when the crowd size equals 10, 20, 30, 40, and 50, respectively. Fig. 11 shows the model

performance with respect to different  $p_{CVI}$  and  $p_{TN}$  based on different crowd sizes. Fig. 11 shows that  $p_{CVI}$  increased accordingly as the crowd size increased, making the crowd consensus model more reliable. In Fig. 11, with the crowd size increased, the curve became sharper. This phenomenon indicated that the  $p_{TN}$  showed increasing stability, which meant that once the crowd size exceeds a certain size, the false positive rate could maintain as 0%. In other words, the crowd consensus model could produce a training image dataset containing no instances that are safe but mistakenly labeled as containing violations. Such dataset is pivotal for minimizing the impact of label noise in the positive training instances that will significantly compromise the performance of the trained advanced computer vision algorithms for workspace safety surveillance [26].

## 5. Discussion

### 5.1. Limitation and future research

First, understanding the underlying cognitive mechanism of the crowd consensus model considering the images contents, safety rules, and cognitive capability of annotators could lead to higher annotation accuracy. The model recognized the fact that the capability of annotators (i.e., passed or failed the quiz) and the difficulty levels of the image-rule pairs (i.e., the image contains violations or not) significantly influence the performance of the annotators. The validation section has proved that the existing crowd consensus model works well even without analyzing and modeling the detailed cognitive process underlying the values of  $p_{ikn}$ . Therefore, it is promising to explore the

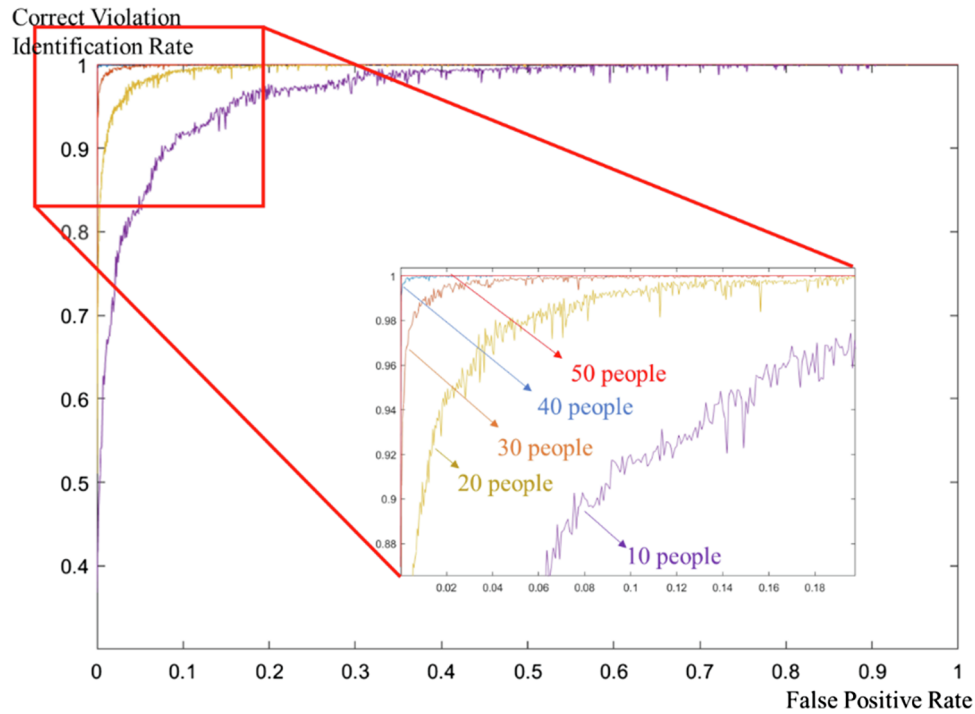


Fig. 11. The trade-off between the posterior false-positive rate (x-axis), which equals to  $1 - po_{TN}$ , and the  $po_{CVI}$  (y-axis) in the experiment results with different crowd size.

underlying cognitive processes of capable and less capable annotators while working on different types of images and rules. Future studies will further investigate and quantify how the complexity of the safety rules, the quality of images, and complexity of the scenes could form different levels of difficulties for hazard detection and trigger different cognitive behaviors of annotators.

Another challenge of the presented new crowdsourcing research is that this approach can only label safety rules violated but could not explicitly label the objects and relations that cause such violations. Labeling rule-related objects and relations is critical for supporting a spatial reasoning algorithm in automatic detection safety rule violations based on object recognition results and spatial arrangement assessment. In the future, the authors plan to examine an improved crowdsourcing game that will collect two annotations on whether the rule is (i) relevant and (ii) violated for each image-rule pair. The game will then present the images, together with relevant or violated rules, to a second crowd, who are tasked to provide evidence of the compliances or violations through highlighting image parts that contain objects relevant to the particular rule.

Third, this study only showed the case that up to one safety violation exists in each workplace image. The challenge in the crowdsourcing game design is that if the annotators designate whether the image violates each safety rule instead of picking one violation in a set of safety rules, the number of questions in the crowdsourcing platform will increase significantly. Therefore, further studies are necessary to focus on the crowd consensus model that supports multi-labeling when it encounters images with multiple violations or particular objects involved in multiple rule violations. One possible approach is to use semantic segmentation methods to segment image into regions representing multiple objects, and then check the spatial relationships between objects in images against multiple safety rules. Some computer vision researchers have been studies the integration of semantic segmentation with natural language processing methods that parse safety rules into object relations [57]. In those studies, semantic segmentation methods segment an image into regions that represent different objects (e.g., the wall, the ground, and the workers). Some objects segmented in the images can form relationships with each other that might violate

safety rules. Then, natural language processing algorithms can automatically process texts of safety rules into “Scene Description Graphs” that specify objects and spatial relationships related to the rule. Overall, segmented images with object information can form a basis for checking object-relations related to specific safety rules. In that way, the integration of semantic segmentation, object detection, and safety rule parsing can examine different combinations of object pairs (e.g., wall vs. worker, ground vs. worker, worker vs. worker) in the image to detect violations of multiple safety rules.

The cost of crowdsourced safety violation labeling will also affect the applicability of the proposed approach. The workload for each person in the crowdsourcing approach may affect the cost as well as the result gathered from the crowd. Future studies will aim at narrowing down the possible safety violation choices to reduce the number of questions to further reduce the cost of crowdsourced safety violation detection. For example, scenario-safety rule compliance checking can identify the safety rules that are impossible to be violated in the images according to the construction task progress and workplace layout.

The efficiency and effectiveness of the training process will greatly influence the performance of crowdsourced safety-rule violation labeling. This study only verified that the current training process could reduce the type-1 majority vote mistakes of the annotations that are difficult to distinguish by the crowd consensus model but did not explore what training method is the most effective in improving the accuracy of annotators. Therefore, another direction for future research is reducing the duration of the training process and improving the quantity/quality of the image annotators’ understanding of the safety rule through the training process. The researchers plan to conduct a series of studies focusing on (1) the maximum amount of safety rules that can be memorized by the anonymous online image annotators through the short training time, (2) the best way to visualize the training material, and (3) and which safety rules usually cause confusion to the image annotators?

The authors will explore the extendibility of the proposed violation labeling method by using larger image sets with various construction scenarios other than elevator installation. These construction scenarios include but are not limited to crane operation, accelerated bridge

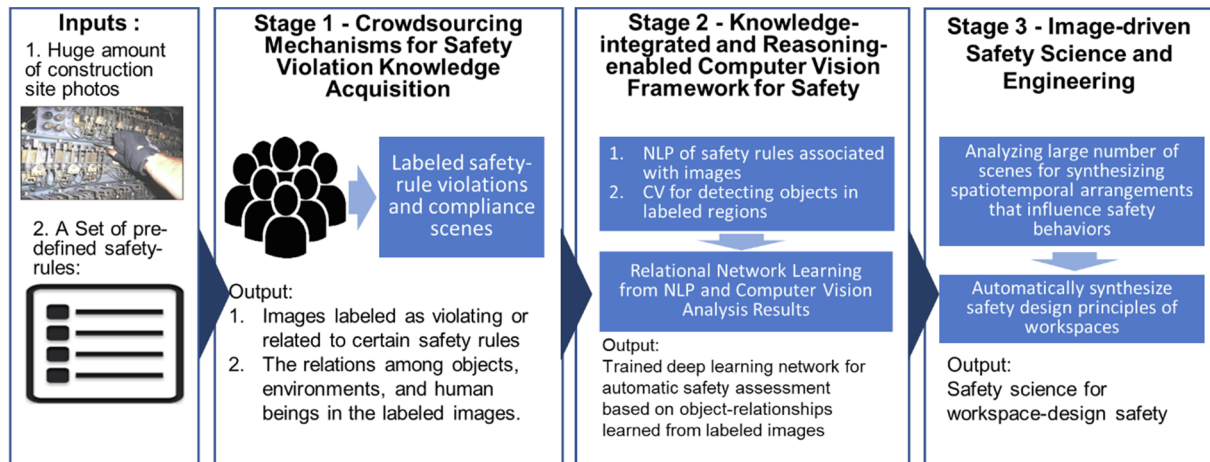


Fig. 12. Envisioned crowdsourced and knowledge-integrated visual reasoning for image-driven safety science.

construction, and modular construction. Currently, computer vision learning research consistently shows that high-quality training dataset leads to higher-quality labeling results [58]. Our algorithm achieved the 95% correct rate of the labeling results. However, the methods still could not reach a 100% correct rate of the labeling results. The authors will further explore methods to find a better solution to improve the developed crowdsourcing approach.

### 5.2. A research roadmap for the automated workspace safety monitoring

The proposed crowdsourcing approach for labeling safety-rule violations on images of complex construction scenes is the first step toward an Automated Workspace Safety Monitoring (AWSM) vision. The overall goal of AWSM is to enable automated and highly-reliable safety monitoring beyond state-of-the-art computer vision methods. The approach is to crowdsource domain-specific knowledge and to integrate the collected knowledge into a knowledge-enabled vision system. Achieving this objective will pave the path toward a real-time spatiotemporal diagnosis of correlations between human-workspace interactions and safety issues. Furthermore, the knowledge-enabled vision system will systematically derive safe and resilient field workflows and spaces. Fig. 12 overviews the proposed framework composed of the following stages:

1. *Crowdsourcing mechanisms for safety knowledge acquisition.* This stage aims at examining crowdsourcing mechanisms augmented by an algorithm that automatically eliminate responses produced by annotators with poor labeling histories. The purpose is to enable acquisition of trustable safety violation assessments based on construction-site images with a variety of contents (objects and environments). The acquired knowledge from each image encodes two types of information: (1) the images labeled as violating certain safety rules, and (2) the relations among objects, environments, and human beings captured in those images.
2. *Knowledge-integrated and reasoning-enabled computer vision framework for safety.* This stage focuses on investigating integrated Natural Language Processing (NLP), and Computer Vision (CV) methods for producing relational networks that can augment a deep-learning approach capable of learning from images containing safety-rule violation and object-relation information to derive which object relationships are related to specific safety rules. The trained deep learning network could identify similar object-relationships in new images that indicate safety violations.
3. *Image-driven safety science and engineering.* This stage focused on applying the developed AWSM framework to examine additional images collected in real workspaces and assist safety engineers in

identifying correlations between spatiotemporal arrangements of workspaces and safety-rule violation rates. Such correlations will define the safety design principles of workspaces. Automated hazard safety-rule compliance monitoring in large amounts of imagery data will assist engineers in advancing safety science – that is, discovering how improper spatiotemporal arrangements of workspaces often setup workers to fail in complying with safety rules. For example, insufficient time and spaces and improper task orders could enforce workers to abandon safety routines or regulations and expose themselves to risks [59–61].

The AWSM framework can ultimately lead to Computer-Aided Workspace Design for workspaces based on large numbers of safety-related spatiotemporal relationships between workers and objects captured in historical and field images or videos. Compared with previous studies [3,5], the AWSM research roadmap aims at enabling computers to learn from human image assessments of workspace images to gain the capability of automatically identifying safety violations and labeling relevant parts in images in real-time.

## 6. Conclusion

This research study shows the capability of using crowdsourcing to label safety-rule violations on images of complex construction workplaces in order to acquire reliable labeling of workspace images about safety violations for training computer-vision-based safety surveillance systems. The goal is to minimize the false alarm rates in the image labels (reporting violations on images having no violations) while keeping an acceptable number of false negatives (reporting images containing violations as “no violations”). Through a short training composed of six illustrations of safety rules, followed by a quiz process on non-professional image annotators, the authors were able to convert the majority mistake from type 1 to type 2. A crowdsourcing type 1 error of the crowdsourced labels is an erroneous bias of the whole population of annotators, while a crowdsourcing type 2 error is a diversity in the labels from annotators without a dominating answer. Type 2 errors usually reflect confusions among multiple possible violations about the image content.

Through the Bayesian consensus model, the authors properly aggregated noisy image labels contributed by ordinary people, filtering out a significant amount of false alarms and acquiring high correct rate of 95%. Compared with the results of majority votes, the proposed crowdsourcing approach, with annotator training and the crowd consensus model, is reliable in discriminating images with and without safety violations. The resultant 0% false-positive and 95% correct rate show the potential of generating datasets about workplace safety

violations, which may advance the AI-based safety monitoring approaches in complex construction workplaces. This research study sheds lights on eliminating construction safety hazards through the integration of human cognitive capability and automation techniques.

### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Acknowledgement

This material is based upon work supported by the U.S. Department of Energy (DOE), the Nuclear Engineering University Program (NEUP) under Award No. DE-NE0008403, and the Natural Science Foundation of China (Award No.: 51578317). Their support is acknowledged and greatly appreciated. All opinions and findings presented are those of authors and do not necessarily reflect the views of DOE and NSFC.

### Appendix

The authors used Google Form to process the crowdsourcing game, which is about annotating the violations of the safety rules in Table 1 in the workplace images shown in Fig. 3. The link to the Google Forms are the following:

- Annotating the violations of safety rule 1-6: <https://forms.gle/gS9qRm71Bb7tHmP9A>
- Annotating the violations of safety rule 7-12: <https://forms.gle/gyhRoukM7JTjQwU6>

### References

- [1] S. Han, S. Lee, A vision-based motion capture and recognition framework for behavior-based safety management, *Autom. Constr.* 35 (2013) 131–141, <https://doi.org/10.1016/j.autcon.2013.05.001>.
- [2] T. Cooke, H. Lingard, A retrospective analysis of work-related deaths in the Australian construction industry, *ARCOM Twenty-Seventh Annu. Conf.* 2011, pp. 279–288.
- [3] M. Liu, S. Han, S. Lee, Tracking-based 3D human skeleton extraction from stereo video camera toward an on-site safety and ergonomic analysis, *Constr. Innov.* 16 (2016) 348–367, <https://doi.org/10.1108/CI-10-2015-0054>.
- [4] S. Tang, M. Golparvar-Fard, Joint reasoning of visual and text data for safety hazard recognition, *Comput. Civ. Eng.* 2017, American Society of Civil Engineers, Reston, VA, 2017, pp. 450–457.
- [5] J. Seo, S. Han, S. Lee, H.H. Kim, Computer vision techniques for construction safety and health monitoring, *Adv. Eng. Informatics* 29 (2015) 239–251, <https://doi.org/10.1016/j.aei.2015.02.001>.
- [6] Microsoft, Microsoft Build 2017, (2017).
- [7] P.-C. Liao, J. Ding, X. Wang, Enhancing cognitive control for improvement of inspection performance: a study of construction safety, *Int. Conf. Eng. Psychol. Cogn. Ergon.* (2016) 311–321.
- [8] D.C. Brabham, Crowdsourcing as a model for problem solving: an introduction and cases, *Converg. Int. J. Res. into New Media Technol.* 14 (2008) 75–90, <https://doi.org/10.1177/1354856507084420>.
- [9] A. Kittur, E.H. Chi, B. Suh, Crowdsourcing user studies with Mechanical Turk, *Proceeding Twenty-Sixth Annu. CHI Conf. Hum. Factors Comput. Syst. - CHI '08*, ACM Press, Florence, Italy, 2008, pp. 453–456.
- [10] B.L. Ranard, Y.P. Ha, Z.F. Meisel, D.A. Asch, S.S. Hill, L.B. Becker, A.K. Seymour, R.M. Merchant, Crowdsourcing—harnessing the masses to advance health and medicine, a systematic review, *J. Gen. Intern. Med.* 29 (2014) 187–203, <https://doi.org/10.1007/s11606-013-2536-8>.
- [11] K. Liu, S.M. Asce, M. Golparvar-fard, A.M. Asce, Crowdsourcing construction activity analysis from jobsite video streams, *J. Constr. Eng. Manage.* 141 (2015) 04015035, [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001010](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001010).
- [12] M.K. Poetz, M. Schreier, The value of crowdsourcing: Can users really compete with professionals in generating new product ideas? *J. Prod. Innov. Manage.* 29 (2012) 245–256, <https://doi.org/10.1111/j.1540-5885.2011.00893.x>.
- [13] D. Zhou, Q. Liu, J.C. Platt, C. Meek, N.B. Shah, Regularized Minimax Conditional Entropy for Crowdsourcing, *ArXiv Prepr. ArXiv1503.07240*. (2015) 1–31.
- [14] Y. Wang, W. Hu, Y. Wu, G. Cao, SmartPhoto: A resource-aware crowdsourcing approach for image sensing with smartphones, *15th ACM Int. Symp. Mob. Ad Hoc Netw. Comput. MobiHoc 2014*, Philadelphia, PA, USA, 2014, pp. 113–122.
- [15] Y. Zhao, Q. Zhu, Evaluation on crowdsourcing research: current status and future direction, *Inf. Syst. Front.* 16 (2014) 417–434, <https://doi.org/10.1007/s10796-012-9350-4>.
- [16] Y. Bachrach, T. Graepel, T. Minka, J. Guiver, How to grade a test without knowing the answers – a Bayesian graphical model for adaptive crowdsourcing and aptitude testing, *Proc. 29th Int. Conf. Mach. Learn.* 2012, pp. 1183–1190.
- [17] G. Radanovic, B. Faltings, R. Jurca, Incentives for effort in crowdsourcing using the peer truth serum, *ACM Trans. Intell. Syst. Technol.* 7 (2016) 1–28, <https://doi.org/10.1145/2856102>.
- [18] A. Burnap, Y. Ren, R. Gerth, G. Papazoglou, R. Gonzalez, P.Y. Papalambros, When crowdsourcing fails: a study of expertise on crowdsourced design evaluation, *J. Mech. Des.* 137 (2015) 031101, <https://doi.org/10.1115/1.4029065>.
- [19] R. Drapeau, L.B. Chilton, D.S. Weld, MicroTalk: using argumentation to improve crowdsourcing accuracy, *Hcomp* (2016) 32–41.
- [20] G. Demartini, D.E. Difallah, P. Cudré-Mauroux, ZenCrowd: leveraging probabilistic reasoning and crowdsourcing techniques for large-scale entity linking, *Proc. 21st Int. Conf. World Wide Web, ACM*, 2012, pp. 469–478.
- [21] B. Frénay, M. Verleysen, Classification in the presence of label noise: a survey, *IEEE Trans. Neural Networks Learn. Syst.* 25 (2014) 845–869.
- [22] T. Xiao, T. Xia, Y. Yang, C. Huang, X. Wang, Learning from massive noisy labeled data for image classification, *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.* 2015, pp. 2691–2699.
- [23] Y. Li, J. Yang, Y. Song, L. Cao, J. Luo, L.-J. Li, Learning from noisy labels with distillation, *Proc. IEEE Int. Conf. Comput. Vis.* 2017, pp. 1910–1918.
- [24] Y. Dgani, H. Greenspan, J. Goldberger, Training a neural network based on unreliable human annotation of medical images, *2018 IEEE 15th Int. Symp. Biomed. Imaging (ISBI 2018)*, IEEE, 2018, pp. 39–42.
- [25] D. Tanaka, D. Ikami, T. Yamasaki, K. Aizawa, Joint optimization framework for learning with noisy labels, *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.* 2018, pp. 5552–5560.
- [26] T. Leung, Y. Song, J. Zhang, Handling label noise in video classification via multiple instance learning, *2011 Int. Conf. Comput. Vis.*, IEEE, 2011, pp. 2056–2063.
- [27] R.E. Levitt, N.M. Samelson, *Construction Safety Management*, John Wiley & Sons, 1993.
- [28] I.G. Awolusi, E.D. Marks, Safety activity analysis framework to evaluate safety performance, *J. Constr. Eng. Manage.* 143 (2016) 05016022, [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001265](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001265).
- [29] H.-H. Wang, F. Boukamp, Ontology-Based representation and reasoning framework for supporting job hazard analysis, *J. Comput. Civ. Eng.* 25 (2011) 442–456, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000125](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000125).
- [30] A.H.M. Rubaiyat, T.T. Toma, M. Kalantari-Khandani, S.A. Rahman, L. Chen, Y. Ye, C.S. Pan, Automatic detection of helmet uses for construction safety, *2016 IEEE/WIC/ACM Int. Conf. Web Intell. Work*, 2016, pp. 135–142.
- [31] SMARTVID.IO, 2017. doi: <http://doi.org/10.5703/1288284316366>.
- [32] P.H. Christensen, W. Jarosz, The path to path-traced movies, *Found. Trends® Comput. Graph. Vis.* 10 (2016) 103–175, <https://doi.org/10.1561/06000000073>.
- [33] I. Arganda-Carreras, S.C. Turaga, D.R. Berger, D. Ciresan, A. Giusti, L.M. Gambardella, J. Schmidhuber, D. Laptev, S. Dwivedi, J.M. Buhmann, T. Liu, M. Seyedhosseini, T. Tasdizen, L. Kamensky, R. Burget, V. Uher, X. Tan, C. Sun, T.D. Pham, E. Bas, M.G. Uzunbas, A. Cardona, J. Schindelin, H.S. Seung, Crowdsourcing the creation of image segmentation algorithms for connectomics, *Front. Neuroanat.* 9 (2015) 1–13, <https://doi.org/10.3389/fnana.2015.00142>.
- [34] J. Deng, J. Krause, L. Fei-Fei, Fine-grained crowdsourcing for fine-grained recognition, *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.* 2013, pp. 580–587, <https://doi.org/10.1109/CVPR.2013.81>.
- [35] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A.C. Berg, L. Fei-Fei, ImageNet large scale visual recognition challenge, *Int. J. Comput. Vis.* 115 (2015) 211–252, <https://doi.org/10.1007/s11263-015-0816-y>.
- [36] J. Deng, W. Dong, R. Socher, L.J. Li, K. Li, L. Fei-Fei, ImageNet: A large-scale hierarchical image database, *2009 IEEE Conf. Comput. Vis. Pattern Recognit.* 2009, pp. 248–255.
- [37] A. Krizhevsky, Learning multiple layers of features from tiny images, *Sci. Dep. Univ. Toronto Tech.* (2009) 1–60 doi: 10.1.1.222.9220.
- [38] M.-W. Park, N. Elsafty, Z. Zhu, Hardhat-wearing detection for enhancing on-site safety of construction workers, *J. Constr. Eng. Manage.* 141 (2015) 04015024, [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0000974](https://doi.org/10.1061/(ASCE)CO.1943-7862.0000974).
- [39] M.-W. Park, I. Brilakis, Continuous localization of construction workers via integration of detection and tracking, *Autom. Constr.* 72 (2016) 129–142, <https://doi.org/10.1016/j.autcon.2016.08.039>.
- [40] M. Memarzadeh, M. Golparvar-Fard, J.C. Nibbles, Automated 2D detection of construction equipment and workers from site video streams using histograms of oriented gradients and colors, *Autom. Constr.* 32 (2013) 24–37, <https://doi.org/10.1016/j.autcon.2012.12.002>.
- [41] C. Koch, K. Georgieva, V. Kasireddy, B. Akinci, P. Fieguth, A review on computer vision based defect detection and condition assessment of concrete and asphalt civil infrastructure, *Adv. Eng. Informatics* 29 (2015) 196–210, <https://doi.org/10.1016/j.aei.2015.01.008>.
- [42] Salary.com, Safety Specialist in the United States, Salary.Com. (n.d.). <https://www1.salary.com/Safety-Specialist-Salaries.html>.
- [43] PayScale.com, Salary for Certification: Occupational Safety and Health Administration (OSHA), (n.d.). [https://www.payscale.com/research/US/Certification=Occupational\\_Safety\\_and\\_Health\\_Administration\\_\(OSHA\)/Salary](https://www.payscale.com/research/US/Certification=Occupational_Safety_and_Health_Administration_(OSHA)/Salary).
- [44] Salary.com, Health and Safety Manager in the United States, Salary.Com. (n.d.). <https://www1.salary.com/Health-and-Safety-Manager-Salaries.html>.
- [45] A. Kovashka, O. Russakovsky, L. Fei-Fei, K. Grauman, Crowdsourcing in computer vision, *Found. Trends® Comput. Graph. Vis.* 10 (2016) 177–243, <https://doi.org/>

- 10.1561/06000000071.
- [46] J. Howe, The rise of crowdsourcing, *Wired Mag.* 14 (2006) 1–5, <https://doi.org/10.1086/599595>.
- [47] Y. Ren, A.E. Bayrak, P.Y. Papalambros, EcoRacer: game-based optimal electric vehicle design and driver control using human players, *J. Mech. Des.* 138 (2016) 061407, , <https://doi.org/10.1115/1.4033426>.
- [48] Y. Sun, C. Dance, How to assure the quality of human computation tasks when majority voting fails, *Work. Comput. Soc. Sci. Wisdom Crowds*, 2011, pp. 1–4.
- [49] C.F. Salk, T. Sturn, L. See, S. Fritz, Limitations of majority agreement in crowd-sourced image interpretation, *Trans. GIS* 21 (2017) 207–223.
- [50] S.-H. Kim, K.-J. Jang, Designing performance analysis and IDEF0 for enterprise modelling in BPR, *Int. J. Prod. Econ.* 76 (2002) 121–133, [https://doi.org/10.1016/S0925-5273\(00\)00154-7](https://doi.org/10.1016/S0925-5273(00)00154-7).
- [51] D.T. Ross, Structured Analysis (SA): A Language for Communicating Ideas, *IEEE Trans. Softw. Eng.* SE-3 (1977) 16–34, <https://doi.org/10.1109/TSE.1977.229900>.
- [52] Amazon Mechanical Turk, (n.d.).
- [53] E. Hollnagel, *Cognitive Reliability and Error Analysis Method (CREAM)*, Elsevier, 1998.
- [54] P.-C. Liao, X. Luo, T. Wang, Y. Su, The mechanism of how design failures cause unsafe behavior: the cognitive reliability and error analysis method (CREAM), *Procedia Eng.* 145 (2016) 715–722.
- [55] 沈祖培, 王遥, 高佳, 人因失误的后果-前因追溯表, (2005).
- [56] elevatorjobsitesafety.com, Elevator Industry Field Employees' Safety Handbook, (2015). [http://www.elevatorjobsitesafety.com/pdf/SafetyHB\\_2015REDRev.pdf](http://www.elevatorjobsitesafety.com/pdf/SafetyHB_2015REDRev.pdf).
- [57] S. Aditya, Y. Yang, C. Baral, Y. Aloimonos, C. Fermüller, Image Understanding using vision and reasoning through Scene Description Graph, *Comput. Vis. Image Underst.* (2018), <https://doi.org/10.1016/j.cviu.2017.12.004>.
- [58] N. Natarajan, I.S. Dhillon, P.K. Ravikumar, A. Tewari, Learning with noisy labels, *Adv. Neural Inf. Process. Syst.* (2013) 1196–1204.
- [59] R.A. Haslam, S.A. Hide, A.G.F. Gibb, D.E. Gyi, T. Pavitt, S. Atkinson, A.R. Duff, Contributing factors in construction accidents, *Appl. Ergon.* 36 (2005) 401–415, <https://doi.org/10.1016/j.apergo.2004.12.002>.
- [60] Y. Li, Y. Hu, B. Xia, M. Skitmore, H. Li, Proactive behavior-based system for controlling safety risks in urban highway construction megaprojects, *Autom. Constr.* 95 (2018) 118–128, <https://doi.org/10.1016/j.autcon.2018.07.021>.
- [61] N.S.N. Yiu, N.N. Sze, D.W.M. Chan, Implementation of safety management systems in Hong Kong construction industry – a safety practitioner's perspective, *J. Safety Res.* 64 (2018) 1–9, <https://doi.org/10.1016/j.jsr.2017.12.011>.